CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 4 printed pages

PART A (80 MARKS)

QUESTION 1

a) Explain the difference between morals, ethics and laws as three distinct forms of guidance for conduct.

(2 marks)

b) Explain six ways in which a professional code of ethics and conduct, such as the ACM code, guides the behaviour of a computing professional.

(6 marks)

QUESTION 2

a) State TWO (2) formal security models that are classified as information flow models, and state the specific concern that the Chinese Wall model is designed to address.

(2 marks)

b) Explain the three main access control structures used to implement access policies, namely the access control matrix, access control lists and capabilities, and for each structure state one advantage or disadvantage associated with its use.

(6 marks)

QUESTION 3

a) State TWO (2) points about software insecurity and buffer overflows as presented in the Software Security lecture.

(2 marks)

b) Explain six (6) aspects of Windows security mechanisms as presented in the Windows Security lecture, covering the role of subjects and tokens, the purpose of Active Directory, the nature of Windows access control, and the risks arising from the complexity of these mechanisms.

(6 marks)

QUESTION 4

a) A biometric system performs its work in two broad stages. List the TWO (2) stages of the biometric authentication process.

(2 marks)

b) Refer to your answer in (a). Discuss biometric mechanisms and technologies as an alternative to passwords, covering the definition of a biometric, the two categories of biometric applications, and the specific technologies with their limitations.

(6 marks)

QUESTION 5

a) State TWO (2) password security controls that can be applied to protect password files and accounts.

(2 marks)

b) State SIX (6) facts about Kerberos and single sign-on (SSO) as authentication mechanisms, covering the motivation for SSO, the design criteria of Kerberos, and the strengths and weaknesses of Kerberos.

(6 marks)

QUESTION 6

a) Explain two characteristics of Trojan horse malware as described in the lecture, focusing on how it behaves and what it does to the host system.

(2 marks)

b) Discuss six malware attack techniques and propagation methods described in the lecture, explaining for each how the technique operates and what impact it has on the infected system or its users.

(6 marks)

QUESTION 7

a) List the four goals of access control and state what a reference monitor does in an access control system.

(2 marks)

b) Describe the Chinese Wall model by stating the concern it is designed to address, explaining its simple security rule and its write-access rule, and explaining how a subject's permitted access changes over time.

(6 marks)

QUESTION 8

a) Explain how a disclosure threat and a usurpation threat each affect the confidentiality, integrity and availability of a system's assets.

(2 marks)

b) State six facts about the Samy worm and the cross-site scripting (XSS) behaviour it exploited, covering how the malicious script was injected, how it spread between users, and the outcome of the incident.

(6 marks)

QUESTION 9

a) Explain how NIDS sensors should be positioned in a network and give one typical location for such a sensor.

(2 marks)

b) Describe six preventive network security mechanisms and, for each, state the protection it provides to an organization's network.

(6 marks)

QUESTION 10

a) Explain two advantages that adopting an organisational code of ethics brings to a computing professional's conduct, and state what may happen to a member who fails to follow such a code.

(2 marks)

b) Discuss the ethical responsibilities of computing professionals and organisations in protecting the privacy of users' personal information, with reference to professional codes of ethics and the constraints under which information specialists must operate.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) State the role of the TPM in a chain of trust, how the chain is built from hardware to software, and how the platform's trusted state is finally established.

(5 marks)

b) Discuss the NGSCB/Nexus architecture by describing its two primary system components and the security functions each performs on the trusted platform.

(5 marks)

QUESTION 2

a) Explain five ways in which intrusion detection systems and related protection mechanisms are used to detect and defend against attacks on computer systems.

(5 marks)

b) Explain five threat classes and countermeasures described in the lecture, covering how threats are defined, how attacks relate to vulnerabilities, and how protection mechanisms and specific defences address these dangers.

(5 marks)

END OF QUESTION PAPER